Yasar Mehmood

CONTACT

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RESEARCH INTERESTS

I am interested in developing practical and robust AI methods for real-world medical applications, with a focus on efficient model adaptation, multimodal learning, and reliable medical AI. Specifically, I aim to develop computationally efficient methods for adapting pretrained models and investigate learning strategies that improve their robustness, generalization, and effectiveness across heterogeneous medical data.

EDUCATION

COMSATS University Islamabad, Lahore Campus, Lahore, Pakistan | 2019 - 2025
PhD in Computer Science

- GPA: 3.96
- Thesis Title: A Framework for the Diagnosis and Prognosis of Brain Tumor Using Deep Learning
- NVidia Academic Hardware Grant

National University of Science and Technology, Islamabad, Pakistan | 2007 - 2011
Master of Science in Computer Software Engineering (MS / 18 years of education)

- GPA: 3.40
- Thesis Title: Incorporating Business Process Change in Information System Design

University of the Punjab, Lahore, Pakistan | 2001 - 2003
Master of Science in Computer Science (16 years of education)

- GPA: 3.37
- FYP Title: Automation of Allama Iqbal Public High School Raiwind Road Kasur

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RESEARCH EXPERIENCE

Visiting Researcher (Remote) | Oct 2025 – Present
Health Futures Center, Arizona State University (ASU), USA

- Ongoing international research collaboration with Prof. Jianming Liang (Arizona State University) on foundation models for medical imaging.
- Conducting systematic, large-scale experimental evaluations of the existing foundation models developed by the lab.
- Actively utilizing ASU's high-performance computing resources (SOL and Phoenix) to support extensive experimentation and analysis.
- Exploring potential extensions of foundation models to additional settings.

HECKTOR 2026 Challenge — Multimodal Head and Neck Cancer Analysis | Apr 2026 – Aug 2026

Information Technology University of the Punjab, Lahore, Pakistan

- Developed a unified framework for head and neck cancer analysis using multimodal FDG-PET/CT imaging and clinical data.
- Developed an adaptive PET/CT fusion approach for tumor segmentation and machine-learning models for TN staging and recurrence-free survival prediction.
- Achieved **5th position** on the held-out test leaderboard; resulting challenge paper accepted for the MICCAI 2026 proceedings. **[Code & Reproducibility]**

Research Supervision | 2025 – TO DATE

Information Technology University of the Punjab, Lahore, Pakistan

Completed

- Waqas Halim (2025 - 2026) - A Sparse Feature Fusion-Based Deep Learning Framework For Adult Glioma Subregion Segmentation
- Mehak Khursheed (2025 - 2026) - Segmentation of Ultra-Low-Field MRI Leveraging Medical Imaging Foundation Models
- Areej Zahid (2025 - 2026) - Automated Quality Assessment of Ultra–Low-Field Pediatric Brain MRI using the Modern Convolutional Neural Network Architectures

Ongoing

- Maryam Pervaiz (2025 - TO DATE) - MedNeXt-Driven Prompt Guidance for Foundation Model in Meningioma Segmentation
- Maham Zulfiqar Hussain (2025 - TO DATE) - Parameter-Efficient Fine-Tuning of the Medical Imaging Foundation Model for Lumbar Spine Segmentation
- Enab Naeem (2025 - TO DATE) - Mitotic Figure Detection in Histopathology Images with Limited Annotated Data

PhD Researcher | Sep 2019 - Apr 2025

COMSATS University Islamabad, Lahore Campus, Lahore, Pakistan

- Conducted research on brain tumor diagnosis with a focus on resource-efficient domain-adaptation, transfer learning, and U-Net architecture
- Published 3 research papers in peer-reviewed journals
- Collaborated with Prof. Anne Martel (Medical Biophysics, University of Toronto) for one year, on the project "Optimal Pre-Training" and gained access to the computational resources through the Digital Research Alliance of Canada as her external collaborator from December 4, 2022, to December 31, 2023

Visiting Researcher (Remote) | Jul 2022 – Jun 2023

College of Health Solutions, Arizona State University, Tempe, Arizona, USA

- Collaborated remotely with Prof. Jianming Liang for one year as a visiting researcher
- Attended weekly lab meetings, contributing to discussions on the project "Optimal Pre-Training," and received mentorship
- Conducted research on determining the optimal fine-tuning strategy (e.g., the number of layers to tune to the target data) based on the characteristics of the pre-training and target datasets

Research Supervision | 2017 – 2023

Virtual University of Pakistan, Lahore, Pakistan

Supervised theses of the following MS in Computer Science (MSCS) students:

- Muhammad Bilal Awan (2023) - Brain Tumor Grade Classification using a Combination of Deep and Handcrafted Features
- Maria Fayyaz (2022) - Classification of Isocitrate Dehydrogenase (IDH) Status in Gliomas using Transfer Learning (View the thesis through the URL: https://vspace.vu.edu.pk/detail.aspx?id=823)
- Haseeb Khalid (2022) - Prediction of 1p/19q Co-deletion Status in Lower Grade Glioma Patients using Convolutional Neural Networks (View the thesis through the URL: https://vspace.vu.edu.pk/detail.aspx?id=750)
- Waqas Ali (2022) - G-Scrum: Gamified Scrum Process Model (View the thesis through the URL: https://vspace.vu.edu.pk/detail.aspx?id=751)
- Sadia Riaz (2019) - Ext-SPIIM: Process Improvement Framework for Global Software Development (View the thesis through the URL: http://vspace.vu.edu.pk/detail.aspx?id=197)

TEACHING EXPERIENCE

Teaching Fellow | Aug 2023 – Aug 2026

Information Technology University of the Punjab, Lahore, Pakistan

Courses Taught:

- Software Engineering
- Probability and Statistics
- Object-Oriented Programming
- Programming Fundamentals

FYPs Supervised:

- AWAAM-AI (Fall 2025 - Spring 2026) - **2nd place among Faculty of Science FYPs**
- Skill Sprint: An AI-Powered Personalized Learning Platform (Fall 2025 - Spring 2026)
- TIXE: A Freelancing Platform (Fall 2025 - Spring 2026)
- RAG-Based Intelligent Course Allocation and Faculty History Management System (Fall 2025 - Spring 2026)
- AirXpert: Customer Support System (Fall 2025 - Spring 2026)
- BeginnerBytes: Gamified Programming Learning Tool (Fall 2023 - Spring 2024)
- Sale Simpli (Fall 2023 - Spring 2024) - A Point of Sale for handheld devices

Visiting Faculty Member | Sep 2022 – Feb 2023

COMSATS University Islamabad, Lahore Campus, Lahore, Pakistan

Courses Taught:

- Web Technologies
- Management Information Systems

Assistant Professor | Dec 2009 – Aug 2022

Virtual University of Pakistan, Lahore, Pakistan

Courses Tutored:

- Software Engineering-I (UG)
- Software Engineering-II (UG)
- Human Computer Interaction (UG)
- Software Quality Assurance (PG)
- Software Design (PG)
- Software Process Improvement (PG)
- Formal Methods for Software Engineering (PG)

Other Responsibilities:

- Managed a team of around 30 e-lecturers/tutors
- Performed the following duties regarding the development of new courses:
  - Developed criteria for the shortlisting of resource persons
  - Searched and contacted the resource persons
  - Reviewed the proposed course outline and detailed contents for the new courses
  - Performed the final QA on the newly developed courses before they were offered
- Supervised theses of 5 MSCS students
- Supervised FYPs of UG students

E-Lecturer | Apr 2006 – Sep 2007

Virtual University of Pakistan, Lahore, Pakistan

Courses Tutored:

- Modern Programming Languages (UG)
- Visual Programming (UG)

Key Responsibilities:

- Developed and evaluated formative (assignments, quizzes, and graded discussions) and summative (mid-term and final-term exams) assessments
- Responded to students' queries on course content, logistics, and assessments
- Collaborated with the faculty to refine the course content and pedagogy

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SELECTED PUBLICATIONS

- **Mehmood, Y.,** Hu, C. A., & Zaidi, M. F. (2026). Deep learning-based segmentation and feature-based clinical prediction for head and neck cancer. In Medical Image Computing and Computer Assisted Intervention – **MICCAI 2026 Workshops and Challenges** (LNCS, Vol. 17271). Springer Nature Switzerland. https://papers.miccai.org/miccai-2026-sat/HECKTOR_003.html
- **Mehmood, Y.**, Bajwa, U. I., & Sun, X. (2022). Resource-efficient domain adaptive pre-training for medical images. **arXiv preprint arXiv:2204.13280**.
- **Mehmood, Y.**, & Bajwa, U. I. (2024). Brain tumor grade classification using the ConvNext architecture. **Digital Health**, 10, 20552076241284920.
- Raza, R., Bajwa, U. I., **Mehmood, Y.**, Anwar, M. W., & Jamal, M. H. (2023). dResU-Net: 3D deep residual U-Net based brain tumor segmentation from multimodal MRI. **Biomedical Signal Processing and Control**, 79, 103861.
- Zulfiqar, F., Bajwa, U. I., & **Mehmood, Y.** (2023). Multi-class classification of brain tumor types from MR images using EfficientNets. **Biomedical Signal Processing and Control**, 84, 104777.

INVITED TUTORIALS

Brain Tumor Segmentation with U-Net Using the BraTS Dataset: A Beginner's Tutorial | Apr 2026

Organizers: RISE-MICCAI

YouTube: https://www.youtube.com/watch?v=ekVPoVwmKK4&t=5941s

Tutorial Notebook: https://tutorial.rise-miccai.org/brats19-segmentation-tutorial/

AWARDS & GRANTS

Travel Grant – MICCAI Society RISE Travel Grant | Sep 27-Oct 1, 2026

Awarded financial support of up to USD 2,500 toward conference registration, membership, lodging, and travel expenses to attend MICCAI 2026 in Strasbourg, France.

Travel Grant – MICCAI Winter School 2025 | Nov 18–19, 2025

Received event sponsorship from Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), in collaboration with the MICCAI 2026 Organizing Committee, to attend the MICCAI Winter School 2025 in Abu Dhabi, UAE.

NVidia Academic Hardware Grant (USD 2400) | Feb 2022

Won the grant for the project "Computationally Efficient Domain-Adaptive Pre-Training"

PROFESSIONAL SERVICE

Reviewer, Machine Learning for Health (ML4H) Symposium - 2024, 2025

OpenReview ID: ~Yasar_Mehmood1

Received mentorship on my reviews in 2024 and 2025

Reviewer, Frontiers of Information Technology (FIT) – 2022, 2023, 2024

Submitted reviews via EasyChair

TECHNICAL SKILLS

Programming Languages

Python, C++, JavaScript, HTML

Tools and Frameworks

PyTorch, Keras, OpenCV, NumPy, Pandas, Matplotlib, Git, Bash, SLURM

Machine/Deep Learning

- Deep Learning Architectures: CNNs, U-Nets, Vision Transformers (ViTs), RNNs, Transformers
- Multimodal Learning: Multimodal Medical Imaging, PET/CT Fusion, Vision-Language Models
- Classical Machine Learning Models: Logistic Regression, Decision Tree, Random Forest, SVM, CatBoost
- Survival Analysis: Random Survival Forest

ADDITIONAL TRAINING

Biology & Medicine

- Introduction to the Biology of Cancer on Coursera (2024). View the certificate at the link: https://coursera.org/share/90d695b9a5453f1c2e739592ecadf28d

Statistics & Research Methodology

- Research Ethics based on the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2: CORE 2022) (June 2023), Certificate # 0000931498
- Basic Statistics on Coursera (2018). View the certificate at the link: https://www.coursera.org/account/accomplishments/certificate/8N6A9XDGXP3J
- Qualitative Research Methods on Coursera (2018). View the certificate at the link: https://www.coursera.org/account/accomplishments/certificate/QMF53LBMMS5P
- Quantitative Methods on Coursera (2018). View the certificate at the link: https://www.coursera.org/account/accomplishments/certificate/MGV36FX73NR7

PROFESSIONAL LINKS

- Google Scholar: https://scholar.google.com/citations?user=_FfQ9IQAAAAJ
- LinkedIn: www.linkedin.com/in/yasar-mehmood-08631a128
- GitHub: https://github.com/mehmoodyasar

REFERENCES

Dr. Jianming Liang (Research Collaborator, Jul 2022 - Jun 2023 and Oct 2025 - Sep 2026)

Professor, Biomedical Informatics, Computer Science, Computer Engineering, Software Engineering, Robotics and Autonomous Systems, and Biomedical Engineering
Health Futures Center, Arizona State University
Email: jianming.liang@asu.edu

Dr. Mudassir Shabbir (Department Chair, Aug 2023 - Mar 2025)

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